Лабораторная работа 2

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Цель работы

Изучить идеологию и применение средств контроля версий. Освоить умения по работе с git.

Выполнение работы

Установка git и gh.

```
mgmizinov@localhost-live:~$ dnf install gh
The requested operation requires superuser privileges. Please log in as a user with elevated rights, or use the "--assumeno" or "--downloadonly" options to run the command without modifying the sys
mgmizinov@localhost-live:~$ sudo dnf install gh
Updating and loading repositories:
Repositories loaded.
Package                                 Arch       Version                                  Repository                Size
Installing:
 gh                                     x86_64     2.87.0-2.fc43                            updates                 37.6 MiB

Transaction Summary:
 Installing:         1 package

Total size of inbound packages is 11 MiB. Need to download 11 MiB.
After this operation, 38 MiB extra will be used (install 38 MiB, remove 0 B).
Is this ok [y/N]: y
[1/1] gh-0:2.87.0-2.fc43.x86_64                                                  100% |   6.1 MiB/s |  11.0 MiB |  00m02s
--------------------------------------------------------------------------------------------------------------------------
[1/1] Total                                                                      100% |   3.7 MiB/s |  11.0 MiB |  00m03s
Running transaction
Importing OpenPGP key 0x31645531:
 UserID     : "Fedora (43) <fedora-43-primary@fedoraproject.org>"
 Fingerprint: C6E7F081CF80E13146676E88829B606631645531
 From       : file:///etc/pki/rpm-gpg/RPM-GPG-KEY-fedora-43-x86_64
Is this ok [y/N]: y
The key was successfully imported.
[1/3] Verify package files                                                       100% |   5.0   B/s |   1.0   B |  00m00s
[2/3] Prepare transaction                                                        100% |   1.0   B/s |   1.0   B |  00m01s
[3/3] Installing gh-0:2.87.0-2.fc43.x86_64                                                                   100% |   2.4 MiB/s |  37.7
Complete!
mgmizinov@localhost-live:~$
```

Рисунок 1 - Установка git и gh

Завершили базовую настройку git

```
mgmizinov@localhost-live:~$ git config --global user.name "Mizinov Mihail"
mgmizinov@localhost-live:~$ git config --global user.email "mgmizinov@gmail.com"
mgmizinov@localhost-live:~$ ^[[200~git config --global core.quotepath false~
bash: git: command not found...
mgmizinov@localhost-live:~$ git config --global core.quotepath false
mgmizinov@localhost-live:~$ git config --global init.defaultBranch master
mgmizinov@localhost-live:~$ git config --global core.autocrlf input
mgmizinov@localhost-live:~$ git config --global core.safecrlf warn
mgmizinov@localhost-live:~$
```

Рисунок 2 - Завершение базовой настройки git

Установили ssh ключи

```
mgmizinov@localhost-live:~$ ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
Enter file in which to save the key (/home/mgmizinov/.ssh/id_rsa):
Created directory '/home/mgmizinov/.ssh'.
Enter passphrase for "/home/mgmizinov/.ssh/id_rsa" (empty for no passphrase):
Enter same passphrase again:
Passphrases do not match.  Try again.
Enter passphrase for "/home/mgmizinov/.ssh/id_rsa" (empty for no passphrase):
Enter same passphrase again:
Passphrases do not match.  Try again.
Enter passphrase for "/home/mgmizinov/.ssh/id_rsa" (empty for no passphrase):
Enter same passphrase again:
Passphrases do not match.  Try again.
Enter passphrase for "/home/mgmizinov/.ssh/id_rsa" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/mgmizinov/.ssh/id_rsa
Your public key has been saved in /home/mgmizinov/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:zLxcuOgw4Hn1Rqj5k/pyg6p3kpki+8LZTH19iPhdpWc mgmizinov@localhost-live
The key's randomart image is:
+---[RSA 4096]----+
|                 |
|                 |
|                 |
|       = .   .   |
|  . . + S o o    |
| . + * B * + E   |
|. B+*.+.B o o    |
|+o*++==o .       |
|+Bo+.=+o         |
+----[SHA256]-----+
mgmizinov@localhost-live:~$
mgmizinov@localhost-live:~$ ssh-keygen -t ed25519

Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/mgmizinov/.ssh/id_ed25519):
Enter passphrase for "/home/mgmizinov/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/mgmizinov/.ssh/id_ed25519
Your public key has been saved in /home/mgmizinov/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:rw+nY5WQmXVeYW6aZ+EMZqvy1MNDhaTrV9PIUTToTq0 mgmizinov@localhost-live
The key's randomart image is:
+--[ED25519 256]--+
|            . ++o|
|           .o.=...|
|         =.o*.B  |
|        =  +.& * |
|        S...B @ .|
|         oo= E   |
```

Рисунок 3 - ssh ключи

Создание ключа pgp

```
mgmizinov@localhost-live:~$ gpg --full-generate-key
gpg (GnuPG) 2.4.8; Copyright (C) 2025 g10 Code GmbH
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

gpg: directory '/home/mgmizinov/.gnupg' created
Please select what kind of key you want:
   (1) RSA and RSA
   (2) DSA and Elgamal
   (3) DSA (sign only)
   (4) RSA (sign only)
   (9) ECC (sign and encrypt) *default*
  (10) ECC (sign only)
  (14) Existing key from card
Your selection? 1
RSA keys may be between 1024 and 4096 bits long.
What keysize do you want? (3072) 4096
Requested keysize is 4096 bits
Please specify how long the key should be valid.
         0 = key does not expire
      <n>  = key expires in n days
      <n>w = key expires in n weeks
      <n>m = key expires in n months
      <n>y = key expires in n years
Key is valid for? (0)
Key does not expire at all
Is this correct? (y/N) y

GnuPG needs to construct a user ID to identify your key.

Real name: mihail
Email address: mgmizinov@gmail.com
Comment:
You selected this USER-ID:
    "mihail <mgmizinov@gmail.com>"

Change (N)ame, (C)omment, (E)mail or (O)kay/(Q)uit?
Change (N)ame, (C)omment, (E)mail or (O)kay/(Q)uit? O
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
gpg: agent_genkey failed: Operation cancelled
Key generation failed: Operation cancelled
mgmizinov@localhost-live:~$
```

Рисунок 4 - pgp ключ

Добавление PGP ключа в GitHub

```
mgmizinov@localhost-live:~$ gpg --list-secret-keys --keyid-format LONG
gpg: checking the trustdb
gpg: marginals needed: 3  completes needed: 1  trust model: pgp
gpg: depth: 0  valid:   1  signed:   0  trust: 0-, 0q, 0n, 0m, 0f, 1u
[keyboxd]
---------
sec   rsa4096/2EA7A991E32B89E1 2026-03-01 [SC]
      4AEA27D7AF68F9B8B3276FAB2EA7A991E32B89E1
uid                 [ultimate] mihail <mgmizinov@gmail.com>
ssb   rsa4096/1CA6812D41C03DD 2026-03-01 [E]

mgmizinov@localhost-live:~$ gpg --armor --export <PGP Fingerprint> | xclip -sel clip
bash: syntax error near unexpected token `|'
mgmizinov@localhost-live:~$ sudo dnf install xclip
Updating and loading repositories:
Repositories loaded.
Package                                                                                                    Arch          Version                                                                    Reposit
Installing:
 xclip                                                                                                     x86_64        0.13-25.git11cba61.fc43                                                    fedora

Transaction Summary:
 Installing:        1 package

Total size of inbound packages is 36 KiB. Need to download 36 KiB.
After this operation, 58 KiB extra will be used (install 58 KiB, remove 0 B).
Is this ok [y/N]: y
[1/1] xclip-0:0.13-25.git11cba61.fc43.x86_64
--------------------
```

```
mgmizinov@localhost-live:~/work/study/2025-2026/Операционные системы$ git clone --recursive git@github.com:mgmizinov/study_2025-2026_os-intro.git os-intro^C
mgmizinov@localhost-live:~/work/study/2025-2026/Операционные системы$ git clone https://github.com/MihailMizinov/study_2025-2026_os-intro.git os-intro
Cloning into 'os-intro'...
remote: Enumerating objects: 41, done.
remote: Counting objects: 100% (41/41), done.
remote: Compressing objects: 100% (39/39), done.
remote: Total 41 (delta 1), reused 28 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (41/41), 25.55 KiB | 493.00 KiB/s, done.
Resolving deltas: 100% (1/1), done.
mgmizinov@localhost-live:~/work/study/2025-2026/Операционные системы$ cd ~/work/study/2022-2023/"Операционные системы"/os-intro
bash: cd: /home/mgmizinov/work/study/2022-2023/Операционные системы/os-intro: No such file or directory
mgmizinov@localhost-live:~/work/study/2025-2026/Операционные системы$ cd ~/work/study/2025-2026/"Операционные системы"/os-intro
mgmizinov@localhost-live:~/work/study/2025-2026/Операционные системы/os-intro$ rm package.json
mgmizinov@localhost-live:~/work/study/2025-2026/Операционные системы/os-intro$ echo os-intro > COURSE
mgmizinov@localhost-live:~/work/study/2025-2026/Операционные системы/os-intro$ make
bash: make: command not found...
mgmizinov@localhost-live:~/work/study/2025-2026/Операционные системы/os-intro$ echo os-intro > COURSE
make
bash: make: command not found...
mgmizinov@localhost-live:~/work/study/2025-2026/Операционные системы/os-intro$ git add .
mgmizinov@localhost-live:~/work/study/2025-2026/Операционные системы/os-intro$ git commit -am 'feat(main): make course structure'
[master 8de2ffb] feat(main): make course structure
 2 files changed, 1 insertion(+), 14 deletions(-)
 delete mode 100644 package.json
mgmizinov@localhost-live:~/work/study/2025-2026/Операционные системы/os-intro$ git push
Username for 'https://github.com': mgmizinov
```

Рисунок 8 - редактирование репозитория

Выводы

Изучил идеологию и применение средств контроля версий. Освоил умения по работе с git.